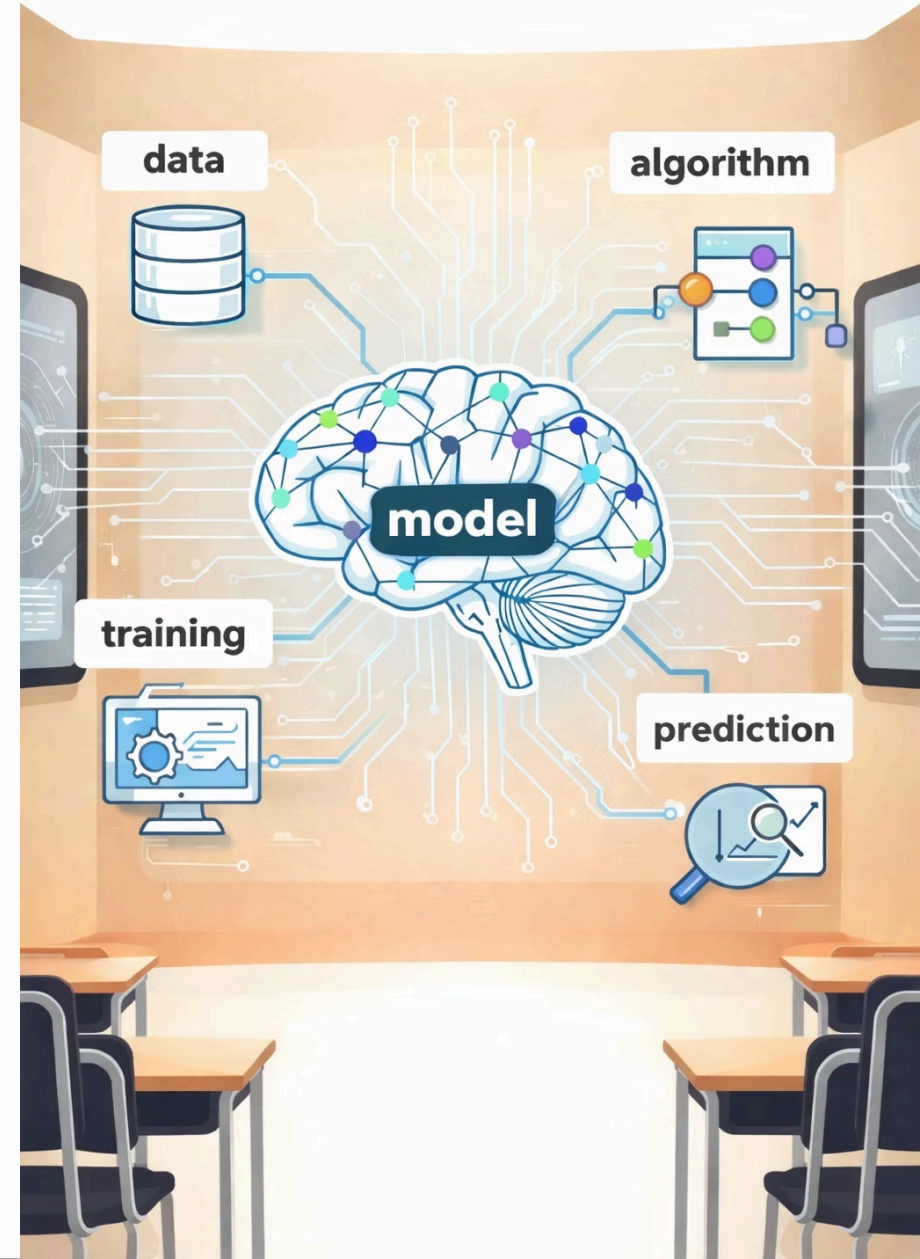


Session 2:

AI Concepts & Terminology

In this session we'll learn the special words that experts use to talk about AI. These are the building blocks you'll use for the rest of your AI journey.

Get ready!



Recap: What We Learned in Session 1

Last time, we discovered some big ideas about AI. Let's refresh our memory:

What is AI?	Artificial Intelligence is a branch of computer science focused on creating machines and programs that can perform tasks that would normally require human intelligence
AI in Daily Life	You already use AI every day - Netflix recommendations, YouTube suggestions, voice assistants like Siri, Google Maps traffic predictions, Spotify playlists, and more
Sense-Think-Act	Every AI system follows this three-stage framework: Sense (gather information), Think (process it), Act (produce an output)
AI vs Human	AI learns differently from humans, has no feelings, lacks common sense, and doesn't truly understand – it finds patterns in data
Myths Busted	AI is not magic, not always correct, not just robots, and doesn't have consciousness
Narrow vs General	All AI that exists today is Narrow AI (designed for one specific task). General AI (machines that can do anything a human can) does not exist yet
AI as Tool	AI assists - YOU create, decide, and are responsible. AI is a tool, not a replacement for human thinking

Today we'll learn the words that helps us talk about *how* those stories were created. What was happening inside the computer when you typed your opening sentence?

Objectives

By the end of this session, you will be able to:

01

AI and Its Subsets

Understand how AI, ML, DL, and GenAI fit together

02

Data - The Fuel of AI

Understand what data means, the different types of data, and why AI needs lots of it

03

Algorithm - The Instructions AI Follows

Explain what an algorithm is and find them in your daily life

04

Training - How AI Learns from Data

Describe what happens when an AI learns from examples

05

Model - The Result of Learning

Explain what an AI model is and what it does

06

Algorithm vs. Model – What's the Difference?

Understand the key difference between these two important ideas

07

Prediction - AI's Smart Guess

Understand what it means when an AI makes a prediction

08

The AI Recipe - Putting It All Together

Connect everything:
Data + Algorithm → Training
→ Model → Prediction

What Do You Already Know About Terms Related to AI?

Before we dive in, let's warm up our brains! Take a moment to think about these questions:

First Thoughts

Have you ever heard words like "algorithm" or "data" before? Where?

Smart Guesses

What do you think "training" means for an AI? How is it different from your training in sports or music?

Burning Questions

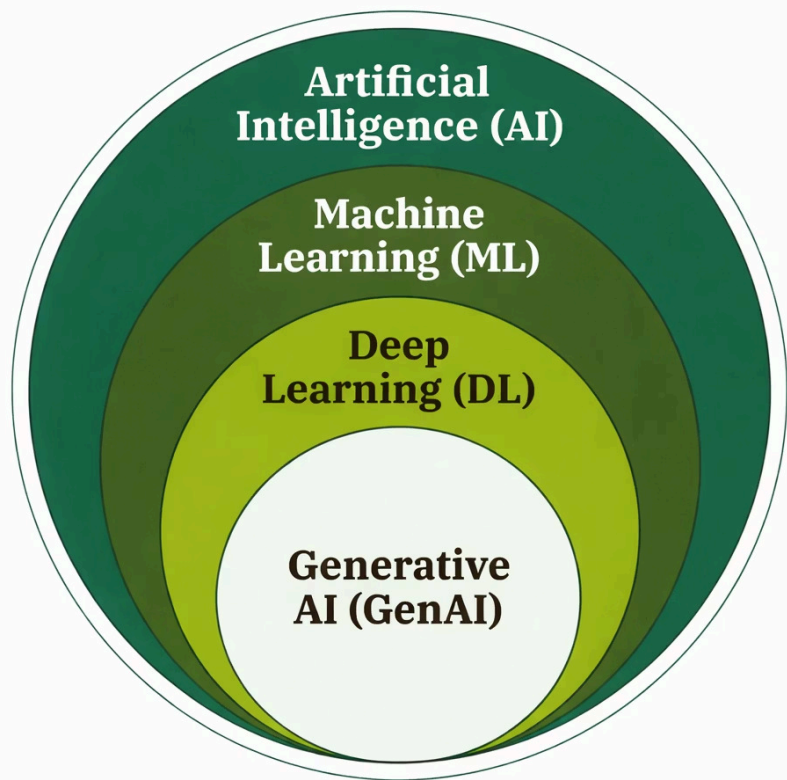
What questions do you have about how AI actually works under the hood?

- ☐ Take a moment to share your thoughts! You might be surprised by how much you already know.



AI and Its Subsets

Before we learn individual words, let's look at the big picture. It's like looking at a map before you start a journey. Understanding how different terms fit together will help you make sense of everything else.



Artificial Intelligence (AI)

The big dream - machines doing smart things

Machine Learning (ML)

A way to achieve AI - computers learning from data

Deep Learning (DL)

A powerful type of ML using brain-like networks

Generative AI (GenAI)

AI that CREATES new things - text, images, music, code

How They Fit Together

Imagine you have :



Large Bin: "AI"

Holds everything related to making machines smart



Smaller Box: "ML"

Holds all the learning-from-data approaches



Even Smaller Case: "DL"

Holds the deep learning methods



Special Compartment: "GenAI"

Holds the AI that creates new things



Your Google Storybook - A Perfect Example

Remember what you created in Session 1? Your story was made by **Generative AI**, which means:

✓ **It's Generative AI**

It created something new

✓ **It's also Deep Learning**

It uses neural networks

✓ **It's also Machine Learning**

It learned from data

✓ **It's also Artificial Intelligence**

It does smart things

One system, all four levels!



Data - The Fuel of AI

Data is information. It's the raw material that AI systems learn from. Without data, AI has nothing to work with - like a car without fuel, a chef with no ingredients, or a painter with no paint.

In computing and AI, **Data** refers to facts, statistics, or pieces of information collected together for reference, analysis, or learning. Data can exist in many forms - numbers, words, images, sounds and serves as the foundation for all machine learning.

Types of Data AI Uses

AI can learn from many different kinds of data. Here are the main types:

Data Type	Examples	What AI Does With It
 Numbers	Temperatures, scores, ages, prices, stock values	Finds patterns, makes predictions, spots trends
 Text	Books, articles, tweets, messages, reviews	Understands language, answers questions, summarizes
 Images	Photos, selfies, drawings, medical X-rays	Recognizes objects, detects faces, identifies things
 Audio	Voice recordings, songs, podcasts, sounds	Understands speech, identifies music, responds to commands
 Video	Movies, TikToks, YouTube clips, security footage	Tracks movement, recognizes actions, detects events
 Sensor Data	GPS location, motion, temperature readings	Predicts traffic, controls robots, tracks fitness

You Create Data Every Single Day!

Every time you use technology, you're creating data. It's happening right now!

Action	What Data Is Created
 Like a video on TikTok	The platform records that you enjoyed this type of content
 Search on Google	Your search terms become data about what people want to know
 Ask Siri a question	Your voice becomes audio data, and your question becomes text data
 Take a photo	That image becomes data - especially if you share it
 Heart a song on Spotify	The platform learns what music you enjoy
 Buy something on Amazon	Your purchase becomes data about shopping patterns
 Share your location	GPS data helps improve traffic predictions for everyone
 Scroll through Instagram	How long you look at each post becomes data about engagement

The Most Important Rule About Data

Garbage In, Garbage Out (GIGO)

If the data you put in is bad, wrong, incomplete, or messy - the AI that comes out will be bad too.

Bad Data	What Happens	Real-World Example
Only photos of brown dogs	AI won't recognize white dogs, black dogs, or spotted dogs	Facial recognition systems trained mostly on light-skinned faces often fail on darker skin tones
Trained only on sunny day photos	AI can't recognize streets, objects, or faces in rain, snow, or fog	Self-driving cars have struggled in weather conditions not in their training data
Only text from one country	AI doesn't understand slang, idioms, or references from other cultures	Chatbots that don't understand regional expressions
Teaching a translation app only with formal textbooks	The AI translates casual conversations into stiff, awkward sentences that sound unnatural	A language app says "Greetings, comrade" instead of "Hey dude"
Training a movie-recommendation AI only on action films	When you ask for a comedy, it keeps suggesting movies with explosions	A streaming service keeps showing you fight scenes even when you want to laugh

Mini Activity : Data Detective

Read each item and write down whether it's: **Number, Text, Image, Audio, or Video**.

Item	Data Type?
A selfie you took with your phone	_____
The lyrics to your favourite song printed on paper	_____
A voice memo of you singing in the shower	_____
The temperature outside right now (72°F)	_____
A TikTok dance video	_____
A text message from your friend saying "See you later!"	_____
A screenshot of your high score in a game	_____
A podcast episode about dinosaurs	_____
A photograph of your family at the beach	_____
Your heartbeat recorded by a smartwatch	_____



Algorithm - The Instructions AI Follows

An **algorithm** is a step-by-step set of instructions to solve a problem or complete a task.

In mathematics and computer science, an **algorithm** is a finite sequence of well-defined instructions for solving a problem or performing a computation.

Algorithms are always:

Clear - each step is precisely defined

Unambiguous - there's no confusion about what to do

Finite - they eventually end after a certain number of steps

Effective - they actually solve the intended problem

Real-Life Algorithms You Already Know

You follow algorithms every single day without even realizing it! Here are some examples:



Getting Ready for School

1. Wake up when alarm rings
2. Brush teeth for 2 minutes
3. Take a shower
4. Dry off and get dressed
5. Eat breakfast
6. Pack backpack with homework and supplies
7. Put on shoes
8. Leave for school



Directions to a Friend's House

1. Turn left on Main Street
2. Go straight for 2 miles
3. Turn right at the gas station
4. It's the third house on the left

Think of It This Way

An algorithm is a **recipe for a computer**.

Recipe	Algorithm
Written for a human cook	Written for a computer
Uses ingredients	Uses data
Produces a dish	Produces a result
Can be followed by anyone	Can be run on any computer

Just like a recipe tells a cook exactly what to do - "mix, then add, then bake at 350°" - an algorithm tells a computer exactly what to do - "compare, then swap, then repeat until sorted." **The computer is the "cook" following the "recipe."**

Mini Activity: Algorithm Hunt

Your task: Find 3 things YOU do step-by-step every day. Write down your algorithm for each one.

Need Ideas?

How you brush your
teeth

How you log into your
favorite game

How you make your
favourite snack or
sandwich

How you get ready for
bed at night

How you solve a math
problem

How you pack your
school bag

How you tie your shoes

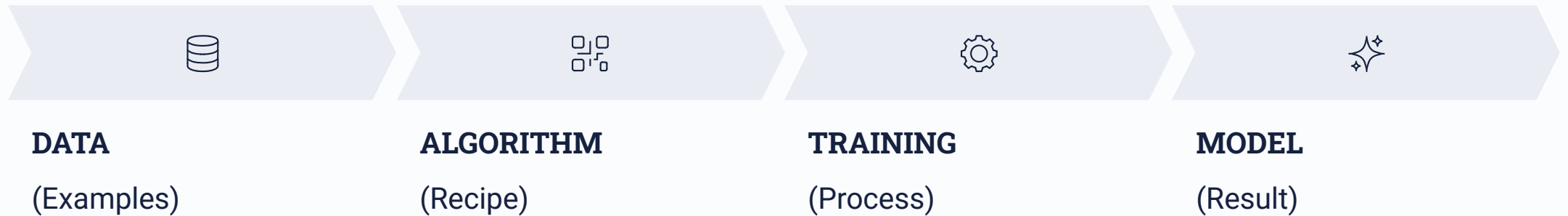
How you navigate to a
new place on your
phone



Training - How AI Learns from Data

Training is the process where an AI system learns from data. It's like studying for a huge test - except the AI studies millions of examples, over and over, until it finds patterns.

How Training Works (Simply)



Data feeds into the algorithm, which goes through training, and produces a model.

The Flashcard Analogy

Imagine you're learning to recognize dogs. You've never seen a dog before. How would you learn?

1 You get flashcards

Each card has a picture on one side. The other side says either "dog" or "not dog." You have thousands of these cards. This is your **DATA** - the examples the AI will learn from.

2 You have a study method

You look at the picture, guess, then flip to check. This is your **ALGORITHM**.

3 You study

You go through thousands of flashcards, again and again. This is **TRAINING**.

4 You become a dog expert!

Now you can look at a NEW picture and correctly say "dog" or "not dog."
You are now a **TRAINED MODEL**.



What Actually Happens Inside AI During Training?

During training, the algorithm goes through a continuous loop:





Imagine you're tuning a guitar string to the perfect pitch:

- 1 — You pluck the string and listen to the note it produces 🎵
- 2 — You compare it to the sound you want (the correct pitch) 👂
- 3 — You adjust the tuning peg just a tiny bit to make it closer 🛠️
- 4 — You pluck again and listen again 🎵
- 5 — You adjust again, even smaller this time 🛠️
- 6 — You repeat this process - pluck, listen, adjust, pluck, listen, adjust until the string plays EXACTLY the right note

❗ Training is exactly like this, but with millions of "strings" inside the AI!

Each "string" is actually a parameter (a number) inside the algorithm. The AI adjusts these numbers by tiny amounts, millions of times, until it can make accurate predictions.

Training Takes... A LOT of Data and Time!

What's Being Trained	How Much Data	How Long It Takes	Note
Simple image classifier (like you'll build in Session 6 using Google Teachable Machine)	50-100 images	A few minutes	Using a pre-built tool - the hard part is already done for you!
ChatGPT / Gemini	Billions of web pages, books, articles – that's petabytes of data!	Months, using thousands of powerful computers running 24/7	Built from scratch by teams of engineers at massive cost
Self-driving car AI	Millions of miles of video footage and sensor data	Years of real and simulated driving	Must learn to handle every possible driving situation



Model - The Result of Learning

A **model** is what an AI system becomes after it has learned from data. It's the trained, ready-to-use version of an algorithm - the final product of the training process.

In machine learning, a **model** is the output of a training algorithm. It's a mathematical representation of patterns learned from data that can be used to make predictions or decisions on new, unseen data. The model captures the relationships, patterns, and rules hidden in the training data.

The Cooking Analogy (Extended)

Cooking Step	AI Step	What It Means
You have a recipe (list of instructions)	You have an algorithm	The method for learning, but not yet skilled
You gather ingredients	You gather data	The raw materials needed
You practice making the dish many times, adjusting as you go	Training happens – the algorithm learns from data	The learning process
You become a skilled chef who knows the recipe by heart	The algorithm becomes a trained model	The final, ready-to-use system
You cook for friends using your skill	The model makes predictions on new data	Using what was learned

❗ The **model** is like a **skilled chef** who has practiced a recipe so many times they can now make it perfectly without looking at the instructions. The recipe (algorithm) is still there, but now there's also skill and knowledge (the model).

Examples of Models You Use Every Day

AI System	What It Was Trained On	The Resulting Model
ChatGPT / Gemini	Billions of web pages, books, articles, and documents from across the internet	A model that can understand language, answer questions, write stories, and have conversations with YOU
Face Unlock on Your Phone	Thousands of photos of YOUR face taken from different angles, in different lighting	A personalized model that knows YOUR face and can recognize you even with glasses, a hat, or a new haircut
Netflix Recommendations	What millions of users watched, liked, skipped, and searched for over many years	A model that predicts what shows and movies YOU might enjoy based on patterns from similar viewers
Spotify / Apple Music	Your listening history, plus what millions of other users listen to	A model that creates personalized playlists like "Discover Weekly" just for YOU
Google Translate	Millions of translated documents – books, websites, and more	A model that can translate between over 100 languages
Google Maps Traffic Prediction	Years of historical traffic data plus real-time GPS data from millions of users	A model that predicts how long your trip will take and suggests the fastest route
Spam Filter in Gmail	Millions of emails marked as "spam" or "not spam" by users	A model that predicts whether a new email is spam and filters it automatically

Key Insight About Models

The model itself is NOT the most important part - the **data** is!

**Same algorithm + Different data
= Different models**

Give the same recipe to two chefs - one gets fresh ingredients, the other gets old ingredients. Very different dishes!

**Different algorithms + Same data
= Different models**

Give two different recipes to chefs using the same ingredients - you'll get different dishes. One might be better than the other for that specific task.

- ① **What data was collected, how it was labeled, what's missing, who was included or excluded - these choices shape EVERYTHING about how the model behaves.**

This is why we say: **The model is not the product, the data pipeline is the product.**

Algorithm vs. Model - What's the Difference?

Algorithm	Model
The recipe	The trained chef
Blank sheet music	A musician who has practiced the piece and can play it beautifully
A workout plan	An athlete who has trained for months and built muscle and skill
General instructions	Specific knowledge
Everyone can have the same algorithm	Each model is unique - based on its specific training data

Most Important Takeaways

Same algorithm + Different data = Different models

Give the same learning method to two different datasets - you'll get two different models. For example, train a face recognition algorithm on photos of mostly one group, and you'll get a model that's bad at recognizing others.

Different algorithms + Same data = Different models

Give two different learning methods to the same dataset - you'll get different models. Some algorithms might work better for certain types of problems.

No model without algorithm AND data

You cannot have a model without BOTH an algorithm (the method) AND data (the examples). An algorithm alone is just instructions. Data alone is just information. Together, through training, they create a model.

Algorithm is general; Model is specific

The algorithm is like a blank template – it can be applied to many problems. The model is the specific result after training on specific data for a specific task.

Hands-On Activity : Explore Gemini

Gemini is a type of AI that was trained on enormous amounts of text from the internet, books, articles, and other sources. It's one of the most advanced AI systems available today.

Gemini can:

- Answer questions on almost any topic
- Write stories, poems, and essays
- Explain complex concepts in simple terms
- Have natural conversations
- Help with creative projects
- Translate between languages
- And much more!



Hands-On Activity : Explore Gemini

Open Gemini (gemini.google.com) and try these three things. Write your answers below.

Task 1: Ask About Data & Training

Ask Gemini:

- "What data were you trained on?"
- "How much text did you learn from?"

What do these answers tell you about the **DATA** and **TRAINING** behind Gemini?

Task 2: Ask for Explanations

Ask Gemini:

- "Explain how a refrigerator works using a simple analogy."
- "Now explain it like I'm 5 years old."

How are they different? What does this tell you about the **MODEL**?

Task 3: Complete These Sentences

Ask Gemini to finish these sentences. Observe what it says.

1. "The first person to walk on the moon was _____"
2. "For my birthday, I hope I get _____"
3. "Once upon a time, in a galaxy far, far away, _____"

What do these examples tell you about how **PREDICTION** works?

What You Discovered

The parts of "**The AI Recipe**" you just experienced:

DATA

ALGORITHM

TRAINING

MODEL

PREDICTION

(You experienced **ALL of them!** The **MODEL** (Gemini) was created through **TRAINING** on massive **DATA** using an **ALGORITHM**, and every response is a **PREDICTION**.)

 **Remember:** Every prediction is based on patterns, not understanding. That means **YOU are still responsible**. Always verify what you learn!

How Gemini Works (Using Our Recipe!)



Data - The Fuel of AI

Billions of web pages, books, articles, and documents from across the internet – that's more text than you could read in a thousand lifetimes!



Algorithm

A deep learning algorithm designed to find patterns in language



Training

Months of training on thousands of powerful computers, processing all that text over and over, adjusting billions of parameters



Model

The Gemini model YOU are talking to right now - a massive neural network with billions of connections



Prediction

Every single word it says is a **prediction** – based on your input and everything it learned during training, it predicts what word should come next

i Every response from Gemini is a prediction! It's not "thinking" or "understanding" in the human sense - it's calculating probabilities and generating the most likely sequence of words.



Prediction - AI's Smart Guess

Prediction is what a trained model does when it encounters new data. It's the model making its best "guess" based on everything it learned during training.

If I say: "The cat sat on the..." You'd probably guess the next word is "**mat**", because that's a common pattern in English. That's a prediction based on patterns you've seen before. AI does exactly this, but with much more complex patterns and billions of examples.

Prediction Is Everywhere!



Netflix

Predicts which show you'll watch next



Google Maps

Predicts how long your trip will take



Gmail

Predicts what word you'll type next



Amazon

Predicts what product you might want



Weather App

Predicts if it'll rain tomorrow



Spotify

Predicts what song you'll love

Training vs. Prediction

Training	Prediction
Learning from examples	Using what was learned
Takes a long time	Happens instantly
Needs lots of data	Needs just one new input
Creates the model	Uses the model

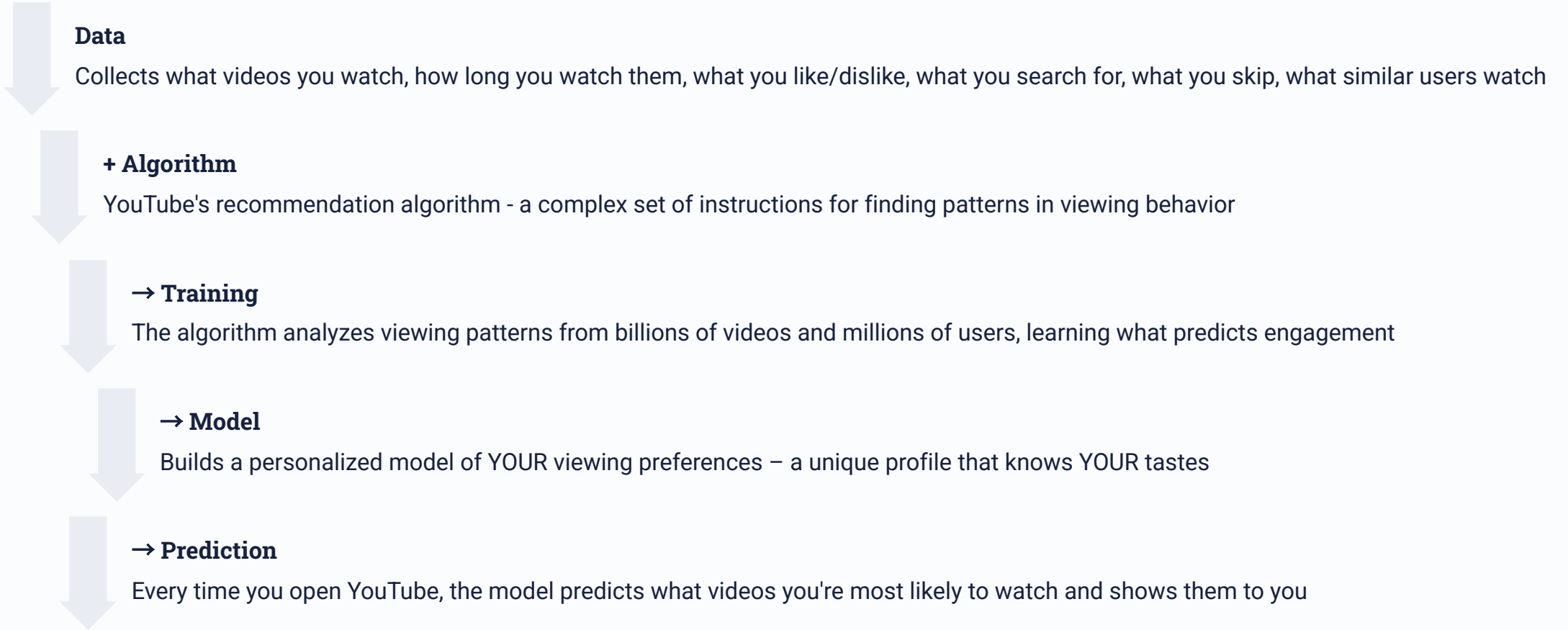
The AI Recipe - Putting It All Together



Data + Algorithm → Training → Model → Prediction

This recipe works for every AI system - from YouTube recommendations to Google Storybook to self-driving cars.

Real Example: YouTube Recommendations



What We Learned Today

	Definition	Example
Data	Information AI learns from	Photos, text, videos, sounds, numbers
Algorithm	Step-by-step instructions for solving a problem	A recipe, directions
Training	The process where AI learns from examples	Studying flashcards, practicing a sport
Model	What an algorithm becomes after training – the trained expert	A skilled chef, a trained athlete, a cat-recognizer
Prediction	The model making a statistical guess on new data	Netflix recommending a show, weather forecast

What We Learned Today

The AI Recipe :

① Data + Algorithm → Training → Model → Prediction

Most Important Takeaways



Data is everything

Garbage in = garbage out. The quality of data determines the quality of AI. Always ask: What data was this trained on?



Training creates the model

Without practice, no skill. A model isn't born smart - it learns from examples. More (good) data generally means better models.



Prediction ≠ Understanding

AI finds patterns, doesn't "know." AI can be confidently wrong. It doesn't understand - it pattern-matches. YOU are responsible for checking.



You can now analyze any AI system

Just ask the five questions: What data? What algorithm? How trained? What model? What predictions?

Homework: AI Detective

Choose ONE AI system you use regularly:

 Netflix / YouTube
recommendations

 Spotify playlists

 Google Maps

 Instagram feed

 Siri / Google Assistant

 Snapchat filters

 Gmail Smart Compose

 Amazon suggestions

Answer these questions:

1. What AI system did you choose?
2. What DATA do you think it was trained on?
3. What do you think the algorithm tries to do? (Example: "Find patterns in what people with similar tastes watch")
4. What does it PREDICT?
5. How does TRAINING help it get better?